

Java

Java when it was initially being developed by SUN was a technology for programming the next generation of smart appliances. We've come a long way since then - and Java has now grown into a programming language that is trusted by legions of programmers across the globe. It's stability and portability have attracted a lot of users, one of the reason why it is the work-horse for enterprise software. The offerings of Java for embedded devices is growing at a tremendous rate. Silicon manufactures are now coming up with embedded Java Virtual Machines eg. the ARM Cortex M-Series which requires only 50K Flash and a few kilobytes of RAM.

The 'write once, run anywhere' approach that Java follows makes it very important and an effective language for embedded systems. As long as there is a JVM for the platform to which it's being deployed, it's guaranteed that your code will run on that machine. Java for embedded systems comes in two flavors:

Java SE Embedded

Java Standard Edition embedded is based on the desktop Java SE. It is designed to be used on systems with at least 32MB of RAM. It can work on Linux ARM, x86, PPC, Windows and Windows CE architectures and platforms.

Java ME Embedded

Java ME embedded is based on the Connected Device Configuration which is a subset of Java Micro Edition. It's designed to be used on systems with at least 8 MB of RAM. It can work on Linux ARM, PPC and MIPS architecture.

Python

Python too can be used on embedded systems. How exactly you can use Python is governed by the limitations of the hardware. Some of the modern embedded devices have enough memory and processor to run a typical-Linux environment. So getting CPython to run on such systems is just a question of cross-compiling and tuning the performance. The devices which can run CPython in are: Gumstix, FIC Neo1973 and Neo FreeRunner and the Telit GSM/GPRS modules.

In this case when we say Python we mean the Python interpreter. For programming the board using python you only need to deploy the Python file to the device where the interpreter will execute the script. In a way this can be considered similar to the JVM.